

Project Progress Report

Project Title

AI Ambulance Intelligence System: Machine Learning-Based Emergency Response Time Prediction and Decision Support

Problem Statement

Emergency Medical Services (EMS) often experience delays due to increasing emergency calls, traffic congestion, and inefficient ambulance allocation. Predicting response times and identifying potential delays using historical EMS data can help optimize emergency operations, reduce response times, and improve patient outcomes.

Objective of the Project

The primary objectives of this project are:

- To analyze large-scale EMS incident data and identify factors affecting ambulance response.
- To build machine learning models for predicting response time, dispatch time, travel time, and emergency demand.
- To classify delayed emergency responses.
- To develop an AI-powered decision support system that assists emergency

Research Gaps

Existing Research	Research Gap	Proposed Solution
Most studies focus only on response time prediction.	Other important EMS stages like dispatch delay and travel time are ignored.	Build multiple predictive models covering the complete EMS workflow.
Existing models use limited spatial and temporal features.	Lack of comprehensive feature engineering.	Incorporate temporal, geographical, and

		frequency-based engineered features.
Many studies use traditional regression methods.	Limited comparison with modern ensemble learning algorithms.	Compare Random Forest, XGBoost, and CatBoost models.
Existing systems mainly provide prediction only.	Lack of an integrated decision-support dashboard.	Develop a Streamlit-based AI Ambulance Intelligence System.
Delay prediction is rarely treated as a classification problem.	Difficulty identifying high-risk delayed incidents.	Implement a Delay Classification model for proactive intervention.

Dataset Name

EMS Incident Dispatch Data

Source:

<https://catalog.data.gov/dataset/ems-incident-dispatch-data>

Dataset Details

Attribute	
Dataset Name	EMS Incident Dispatch Data
Original Dataset Size	~6 GB
Processed Dataset Used	100,000 records
Additional Recent Dataset	Last 10,000 records (2024–2025)
Number of Features	11 original features (expanded)

Important Features

- Incident Date & Time
- Initial Call Type
- Final Call Type

- Borough
- ZIP Code
- Dispatch Area
- Dispatch Response Time
- Incident Response Time
- Travel Time
- Hospital Arrival Time
- Incident Close Time

Data Preprocessing Performed

The following preprocessing steps were carried out:

- Removed duplicate records
- Handled missing values
- Converted datetime columns into proper timestamp format
- Extracted temporal features:
 - Hour
 - Day
 - Month
 - Weekday
 - Year
- Label encoded categorical variables
- Frequency encoding for:
 - ZIP Code
 - Borough
 - Call Types
 - Dispatch Area
- Created new engineered features including:

- Dispatch Delay
- Incident Duration
- Peak Hour
- Rush Hour
- Office Hours
- Night Shift
- Weekend Indicator
- Call-Borough Mapping
- ZIP-Hour Mapping

Machine Learning Models Implemented

The project consists of five predictive models:

Model	Task
Response Time Prediction	Regression
Dispatch Time Prediction	Regression
Travel Time Prediction	Regression
Delay Prediction	Classification

Emergency Demand Forecasting

For regression tasks, the following algorithms were implemented:

- Random Forest Regressor
- XGBoost Regressor
- CatBoost Regressor

For classification:

- Random Forest Classifier

Best Model Selected

CatBoost Regressor

It achieved the best overall balance between prediction accuracy and generalization on EMS response time prediction.

Performance Metrics Achieved

Response Time Prediction

Model	MAE	RMSE
Random Forest	330.01	713.03
XGBoost	320.24	694.43
CatBoost	318.74	686.17

Classification Metrics

The Delay Prediction model is evaluated using:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC

Current Findings / Observations

- Temporal features significantly improve prediction accuracy.
- Frequency encoding performs better than simple label encoding for categorical variables.
- CatBoost handles categorical and numerical data effectively with minimal preprocessing.
- Engineered features such as Dispatch Delay and Incident Duration contribute significantly to model performance.
- Predicting multiple EMS stages provides a more comprehensive emergency response analysis than focusing solely on response time.

Challenges Faced

- Large dataset size (~6 GB) required sampling and optimized processing.
- Missing values and inconsistent datetime formats required extensive preprocessing.
- Model compatibility issues arose due to differences in Python and library versions between Google Colab and the local environment.

- High-cardinality categorical variables increased preprocessing complexity.
- Time-series forecasting required additional feature engineering and aggregation techniques.

Next Steps / Planned Improvements

- Live ambulance tracking using GPS
- Real-time traffic integration
- Deep Learning based prediction
- Multi-city deployment
- Cloud deployment
- Hospital availability prediction
- Route optimization